

Learned model — RATIONALE (the *why* behind every estimator)

The conceptual companion to `LEARNED_MODEL_METHODS.md`. METHODS is the **spec** — notation, objective, formula, one line of “what it computes” — deliberately kept clean. This doc holds the **reasoning** that does not belong in a formula sheet: *why this construction and not the naive one, what breaks otherwise, what the estimator is really identifying, and how to read its output honestly*. Section numbers mirror METHODS exactly (§0–§17), so a reader can hold the two side by side.

- **What is the formula?** → `LEARNED_MODEL_METHODS.md`
 - **Which estimator for which job, and chosen over what?** → `LEARNED_MODEL_ESTIMATOR_MAP.md`
 - **Does it actually work (numbers, positive controls, honest negatives)?** → `LEARNED_MODEL_VALIDATION.md`
 - **Why this design at all (decision log)?** → `LEARNED_MODEL_DESIGN_RESPONSE.md`
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The one idea that organizes everything

Every estimator here answers a question about the map *miRNA-arm abundance* → *target mRNA*, and the whole design turns on **matching the estimator to the estimand’s robustness**. Two questions that look identical have opposite stability under the central nuisance — **seed-family collinearity** (same-seed arms move together):

- **Aggregate / selection / prediction** questions (does coupling exist? does the model beat abundance? does a new edge enter?) read the *bundle* $X \cdot M$ or an in/out decision. Collinearity inflates *coefficient* variance but **not** fitted-value variance — the aggregate is stable even when the individual weights are not. → estimated with the **model** (adaptive lasso).
- **Coefficient-read / attribution** questions (how much does *this* arm contribute? how does its weight shift across states?) read the individual $M(m, g)$ — precisely the object collinearity destabilizes (single-fit bootstrap correlation ≈ 0.03). → estimated with **bagged NNLS on a fixed family support**.

If you ever feel two sections are “doing the same thing twice,” this is why they are not: they read different functionals of the same fit, with different robustness, so they need different machinery.

§0 Notation & identification — why residual space, why POOLED-HE

Why everything lives in the C-residual space ($I - P_C$). We never want to know “does a high-miRNA patient have a low target” — that correlation is polluted by purity, copy number, and proliferation, all of which move miRNA and mRNA jointly. We want the *within-stratum* question: holding those fixed, does more of this arm mean less target. Residualizing **both** predictor and target on C (Frisch-Waugh-Lovell) makes every downstream coupling a **partial** association by construction, so no later step has to “remember” to adjust — the confounders are already gone from the coordinates.

Why POOLED-HE is the candidate set, not all predicted edges. The universe of *predicted* miRNA→target pairs is enormous and mostly false; feeding it to the model would drown real edges in selection noise. We restrict candidacy to edges with **experimental** support (miR-TarBase high-evidence $\cup$ TarBase-v9 low-throughput functional). This is a *prior on existence*,

imposed before any TCGA data is seen — which is what keeps the model non-circular. The migration from miRTarBase-HE-only to POOLED-HE was checked to be gate-neutral on anchors ($\Delta p \leq 0.003$): it adds coverage without moving the headline.

§1 Confounder block C — why each term, and the composition trap

The subtle confounder is **cell composition**, and it is a trap because it hides in *both* modalities. A tumour that is 80% epithelial vs 40% epithelial differs in bulk miRNA and bulk mRNA for reasons that have nothing to do with regulation. So C is built to strip composition **without** stripping the biology:

- CPE (purity) and target_cn (the gene’s own copy number) remove the two dosage confounders that would otherwise fake a miRNA→mRNA link.
- mal_prolif is the delicate one: proliferation drives both, but “proliferation” as measured in bulk is partly *just* the epithelial fraction. So we residualize the proliferation metagene on the Cancer-Epithelial fraction first — leaving **per-malignant-cell** proliferation, the biological part, and discarding the composition part.
- **Why Cancer-Epithelial is excluded from the deconv block** (the deliberate asymmetry): the target is *expressed in* the epithelial compartment, so conditioning on that compartment’s fraction over-controls — it regresses away the very signal we want. We strip the *non-malignant* fractions (they are pure nuisance) but keep the compartment the biology lives in.

Why the deconv default flips by task. Gate/coupling runs with **core C only**: we are asking “is there coupling at all,” and a leaner C has more power. The canonical *attribution* runs with **deconv ON**: when we name a driver we want a **cell-intrinsic** weight, not one an analyst can dismiss as stroma — so we pay the power cost to buy the stronger claim. Same model, different burden of proof, different C.

Why batch is post-fit only. An in-lasso ComBat-style batch correction removes biological heterogeneity along with plate effects (it cannot tell them apart inside the fit). So batch conditions the **significance test** (oof_gate’s ρ computation) but never the **model** — we correct the ruler, not the object.

§2 The prior w — inverse penalty, not a location (and where spike-and-slab would go)

The evidence prior is the project’s most misread object, so state it plainly: **w_m is an inverse penalty, not a coefficient**. A strong-evidence edge is not pushed toward a large weight; it is made **harder to zero out**. Formally $\lambda_m = \alpha / w_m$, so the prior sets *ordering and selection* (who survives shrinkage) while the **data set magnitude** (§3). This is why the source of w is empirically non-critical (ledger $\approx$ evidence $\approx$ fused within 0.005 on the gate): the prior only has to get the *ranking* roughly right, and any reasonable evidence fusion does.

Why fused/deduped magnitude and not binary inclusion. Binary “is it curated” throws away that some edges have one qPCR and some have ten reporter assays. $w_m = \sum_{\text{class}} w_{\text{class}} \cdot \log_{1p}(\# \text{distinct PMIDs})$ keeps that gradient, deduplicated across databases so a paper both DBs cite counts once, and log1p so the tenth PMID matters less than the second. w_{class} grades by assay *directness* (reporter > western > qPCR > CLIP) because a luciferase construct is closer to “this arm represses this gene” than a binding footprint.

Where the spike-and-slab fits — and why the lasso is what ships. The parked Bayesian design (LEARNED_MODEL_DESIGN_RESPONSE §D/E) replaces the lasso’s *implicit* L1 selection with an **explicit, evidence-graded inclusion indicator** $z_{\{g,m\}} \sim \text{Bernoulli}(\pi_{\{g,m\}})$ — a true “does this edge exist” random variable, separate from magnitude. It is now **built and benchmarked** (learned/spike_slab.py; §2a below). Verdict: it is competitive and recovers the

curated drivers, but it does **not** beat the adaptive lasso’s soft-L1 selection on out-of-fold coupling — which is exactly why the lasso ships and the spike-and-slab stays the documented alternative rather than the production model.

§2a Spike-and-slab benchmark — the honest head-to-head

The model: $r = X\beta + \varepsilon$, $\beta_m = z_m \cdot \theta_m$, $z_m \sim \text{Bernoulli}(\pi_m)$ with π_m monotone in the evidence prior, $\theta_m \sim N^+(\theta, \sigma^2)$ (half-normal slab $\rightarrow$ keeps the $M \geq \theta$ repression box). Evidence enters **only** through inclusion π , so the benchmark isolates the design’s actual claim. Estimated by componentwise SSVS Gibbs (`spike_slab.gibbs_ss`); the gate reads the posterior-mean coefficient $M_m = E[z_m \cdot \theta_m]$ and PIP $E[z_m]$, scored on the **same folds / C / family predictors** as the lasso.

- **Hub panel:** beats abundance **4/5**, recovers curated drivers at $\text{PIP} \approx 1.0$ (ZEB1 miR-200/429, ESR1 miR-18, PTEN miR-17~92), but matches/beats the lasso only **2/5** (mean $\rho -0.313$ vs lasso -0.351). It *wins* where the true support is a tight cooperative family (ZEB1) and *loses* where many weak arms add up (PTEN, CDKN1A) — the lasso’s graded shrinkage keeps more of that diffuse signal than a hard in/out indicator.
- **Genome-wide sample (107 genes):** the gap holds and is decisive — spike-and-slab is FDR-significant on **35%** of genes vs the lasso’s **45%**, mean ρ **-0.078 vs -0.125** , and matches/beats the lasso on only **21%** of genes. (Run: `spike_slab.gate_fdr(limit=120)` $\rightarrow$ `output/learned/gate_fdr_spike_slab.tsv`; lasso ρ on this sample = -0.125 , matching the published genome-wide -0.123 , so the sample is representative.)
- **Takeaway:** an explicit inclusion prior buys **interpretability** (a real PIP per edge) but not **coupling** over the lasso. It is the right tool when the question is “does this edge exist, with a posterior probability” and the wrong tool when the question is “predict the target as well as possible.”

PIP = Posterior Inclusion Probability = $E[z_m]$, the fraction of post-burn Gibbs samples in which edge m was included. It is the model’s “probability this edge is a real regulator, given data + prior”; $\text{PIP} = 1.00$ means every sample kept it (e.g. ESR1 miR-18, PTEN miR-17~92). This is the object the lasso *cannot* give — a lasso coefficient is either zero or not, with no calibrated probability attached.

Where it wins, and by how much. The spike-and-slab beats the lasso exactly where the true support is a tight cooperative seed-family and hard inclusion is *correct*: **ZEB1 $\rho -0.465$ vs lasso -0.438 (~ 0.03 deeper, $\sim 6\%$)**, GATA3 -0.204 vs -0.200 . It loses on the diffuse genes where many weak arms each contribute a little (PTEN, CDKN1A) — a Bernoulli in/out discards signal the lasso’s graded shrinkage keeps.

§2b Tuning tried (and why none of it beats the lasso globally)

Two principled modifications were run head-to-head on the hub panel: 1. **Full evidence-graded slab** (`priors.edge_priors  $\rightarrow \pi$` from the ledger + a slab `scale 1+gain $\cdot \mu \cdot \tau$` widened by `scanMiR` biochemical affinity μ and evidence depth τ ; `spike_slab.fit_gene_ss(use_priors=True)`): **a wash** — mean $\rho -0.309$ vs the uniform-slab -0.313 . This is the same lesson as §2/§3 for the lasso: *the prior sets ordering, not magnitude* — a fancier magnitude prior does not move coupling. (The scale prior is deliberately location-at-0, so it never biases the coefficient up; it only loosens shrinkage.) 2. **Base inclusion rate p_0** (the real lever): the diffuse genes want *more* inclusion — PTEN improves monotonically (p_0 $0.3 \rightarrow 0.5 \rightarrow 0.7$ gives $\rho -0.273 \rightarrow -0.311 \rightarrow -0.393$, finally *beating* lasso -0.334), but CDKN1A peaks at $p_0=0.5$ (-0.193) then falls. So the optimum is **per-gene**, and pushing p_0 up globally turns the spike-and-slab into an approximate ridge — surrendering the sparse, interpretable PIP that was its only advantage over the lasso. There is no single global p_0 that beats the lasso across genes. The one modification that *could* help — an empirical-Bayes / per-gene adaptive p_0 — is essentially

re-deriving the lasso’s continuous shrinkage in Bayesian clothing, which is why it is not pursued.

§2c Learned shrinkage TOWARD the prior — empirical Bayes (the τ^2 question)

A distinct and principled idea: instead of using the curated prior as an inverse *penalty* (ordering) and *guessing* the shrinkage strength α , make the prior a *location* the coefficients shrink toward and let the **data learn how hard** via a between-effect variance τ^2 — small τ^2 where the prior agrees with the data (shrink hard, trust the prior), large τ^2 where it disagrees (shrink little, trust the data). Built as `eb_shrink.py`: per gene $\beta_m \sim N(c \cdot \hat{w}_m, \tau^2)$, $\hat{w}$ = unit-normalised curated weight, c and τ^2 learned by conjugate Gibbs; c = how much the prior’s *direction* is used at all. **This is genuinely different from §10** (which learns τ^2 but anchors to a *data-estimated* cross-gene mean μ_m , not the curated prior). Result on the hub panel — the data’s own verdict on the prior:

gene	ρ_{eb}	ρ_{lasso}	learned τ^2	c	τ^2/c^2
PTEN	-0.204	-0.334	0.033	0.011	273
GATA3	-0.185	-0.200	2.365	-0.006	56870
ESR1	-0.473	-0.536	0.175	0.270	2.4
ZEB1	-0.596	-0.438	0.110	0.039	71
CDKN1A	-0.260	-0.246	0.062	0.012	431

- **The learned τ^2 CONFIRMS the design empirically.** τ^2/c^2 (data dispersion $\div$ prior-anchored part) is 71–57000 in 4/5 genes — the data-estimated between-effect variance *swamps* the prior-anchored component, so the model chooses to shrink little toward the prior and let the data set magnitude. “Prior sets ordering, not magnitude” is thus **derived from the data** (τ^2 learned large), not assumed. ESR1 (ratio 2.4, $c=0.27$) is the one gene where the prior magnitude genuinely pulls.
- **Coupling is a wash** (mean ρ -0.344 vs lasso -0.351): learning the anchor strength doesn’t beat guessing α , because coupling is insensitive to shrinkage strength (the flat α -sweep, §3). Learning τ^2 *justifies* the guess rather than improving on it — real methodological value (removes the “ α is arbitrary” objection), zero headline movement.
- **Side-finding:** where EB *wins* (ZEB1 -0.596 vs -0.438, large) it is **not** the prior ($c \approx 0$) — it is that a **Gaussian ridge beats L1 on tight cooperative families** (keeps all correlated members instead of selecting one). Same direction as the spike-slab ZEB1 win, larger. This lead was chased at full scale — §2d.

§2d Learned- τ^2 ridge vs the lasso — GENOME-WIDE (the one thing that moved the needle)

Chasing §2c’s ZEB1 lead: is a **learned- τ^2 non-negative ridge** (dense, shrinkage strength learned by the between-effect variance ν^2 , repress-box intact) a better *coupling* estimator than the shipped lasso? Built as `shrinkage_compare.py` (all estimators in ONE leak-free harness — train-only residualisation, adaptive w-scaling, raw-X aggregate — differing only in penalty + λ -source). Disentangling the two axes the ZEB1 teaser conflated (penalty type vs the $M \geq 0$ box):

- **Penalty type is inert; λ -SOURCE is everything.** `nnridge_cv` (L2, λ by CV) $\approx$ `nnlasso` (-0.1525 vs -0.1518 genome-wide) — swapping L1→L2 does nothing. But `ebbridge_nn` (L2, λ learned by ν^2) beats the lasso: **mean -0.1683 vs -0.1518, wins 379/655 multi-regulator genes (58%), Wilcoxon $p=9 \times 10^{-16}$** . So it is the **learned τ^2** , not L2, and CV was the wrong stand-in (it optimises prediction MSE, which over-shrinks for coupling). Robust to the ν^2 hyperprior (InvGamma(a0,b0) swept 0.01→20: PTEN

-0.388↔-0.395, always beating lasso -0.219) — the win is NOT a tuned knob (§2c’s “n»p pins ν^2 ”). MacKay type-II ML (ebridge_unc) is the CV-free version and agrees.

- **Two honest caveats full scale exposes (n=5 hid both):** (1) **tail-driven** — mean Δ -0.0165 but **median Δ -0.003**; the typical gene barely moves, the lift is concentrated in genes with MANY regulators (CCL5, MET n=24, E2F2, cytokines/MMPs) where lasso’s sparse selection discards signal. (2) The **cooperative-family/collinearity hypothesis FAILS at scale** — Spearman(collin, Δ)=-0.074, non-monotonic across tertiles. The mechanism is **regulator-count, not collinearity**: at n»p, dense learned- τ^2 shrinkage beats sparse selection *when there are many arms to select among*. The ZEB1 collinearity story was small-n.
- **Verdict:** learned- τ^2 NN-ridge is a genuinely better *coupling* estimator (significant, biology-intact), but its edge is concentrated in many-regulator genes, and it is **dense** (no selection) — so it is a better fit for the *prediction/coupling* job, not a drop-in for the lasso’s selection/discovery role (match estimator to estimand). out-put/learned/shrinkage_compare.tsv.

§2e Learned- τ^2 for ATTRIBUTION — EB posterior vs bagged NNLS (attribution_eb.py)

Can the same learned- τ^2 model also serve the *attribution* estimand (the per-edge weight bagged-NNLS reads, §5), and does it add calibrated uncertainty? Per gene, the non-negative learned- τ^2 **posterior** (half-normal Gibbs, $\pi \equiv 1 \rightarrow$ _gibbs_posterior mean+sd) vs bagged NNLS, on the SAME z-scored family predictors: - **Stabilises as well as bagging:** reproducibility EB **0.996** $\geq$ NNLS 0.983 (single-lasso ≈ 0.03); agreement ρ 0.844. - **Recovers curated drivers** via the identified set $|\text{mean/sd}| > 2$: PTEN miR-21/103/181/182 (4/4), GATA3 miR-27, ESR1 miR-18 — and **correctly declines** the §9-unidentified ones (ESR1 miR-221/222; ZEB1’s miR-200 split, $z \approx 1.7$). The EB $|z| > 2$ line **reproduces §9’s conditioned-partial identifiability verdict for free** — one fit gives weight + identifiability, consolidating two estimators. - **Caveat:** the raw EB *mean* is biased UP for un-informed edges (the half-normal prior has a positive mean $\rightarrow$ zero-signal families drift to it, not to 0 like NNLS). Must read through the $|z| > 2$ filter, or switch on the evidence-graded inclusion π (spike-and-slab) to zero them at source. So “EB mean” is not a naked drop-in for “NNLS weight”; the deployable form is **NNLS-style point estimate + EB posterior sd**. - **Stabilisation is the SHRINKAGE, not the prior** (works prior-free); the prior’s only distinct attribution role — tie-breaking collinear members — is deliberately NOT used (it would undo §9’s honest “unidentified”).

§2f Combining τ^2 with the LASSO — the Bayesian lasso (redundant, and why)

Natural idea: learned- τ^2 is not ridge-by-definition (§1’s L1/L2 point) — a **Laplace** prior gives the **Bayesian lasso** (Park & Casella), which *should* keep sparse selection AND learn the shrinkage AND give a posterior. Built (spike_slab_gibbs_blasso, non-negative: per-edge $\tau_j^2 \sim \text{Exp}(\lambda^2/2)$, global λ^2 learned by Gamma) and wired into both harnesses. Result — **it collapses to the ridge on all three jobs**: - **Coupling:** hub mean -0.363 $\approx$ ebridge_nn -0.362 (no gain). - **Attribution:** reproducibility 0.996, agreement-with-NNLS 0.87, driver recovery via $|z| > 2$ — all identical to the ridge. Does **not** fix the un-informed-edge mean bias — marginally worse (PTEN spurious miR-718 bl_mean 0.170 > ridge 0.149: the Laplace’s heavy tail lets a signal-free edge draw a large local $\tau_j^2 \rightarrow$ larger truncated positive mean). $|z| > 2$ kills it either way (bl_z 0.99). - **Why:** we read the **posterior MEAN**, which for a Laplace prior is **not sparse** — only the MODE (MAP) is. Bayesian model-averaging erases the L1 sparsity, so blasso-mean $\approx$ ridge-mean on both aggregate and per-edge. **Genuine Bayesian selection needs an explicit inclusion indicator (the spike-and-slab, §2a) — not a Laplace prior — and that estimator LOSES on coupling** (sparse hurts the aggregate at n»p).

The fundamental split this exposes, and its reconciliation: - **coupling** wants **DENSE** (ridge / blasso-mean / learned- τ^2) — selection hurts; - **attribution** wants a **CLEAN/sparse**

point estimate — but the sparse spike-and-slab hurts coupling; - **the $|z|>2$ identified-set reconciles them** — read the *dense* learned- τ^2 model through the identifiability filter $\rightarrow$ clean attribution (drivers recovered, §9 reproduced) *from* the same dense model that wins coupling.

So the one unified estimator is the learned- τ^2 NON-NEGATIVE model (ridge; blasso is equivalent, adds nothing), with three readouts: **aggregate $\rightarrow$ coupling (§2d), posterior mean $\pm$ sd $\rightarrow$ attribution (§5), $|z|>2 \rightarrow$ identifiability (§9).** Combining τ^2 with the lasso is *redundant*, not additive. See §2g for the one genuine tension this leaves.

§2g Adding the INCLUSION indicator (spike-slab-lasso) — the tension that has no single-fit fix

The last combination: inclusion $z \sim \text{Bernoulli}(\pi_{\text{evidence}})$ + Laplace slab + learned λ^2 — genuine selection AND Bayesian shrinkage (`spike_slab._gibbs_ss_blasso`), tested for BOTH coupling and attribution. - **Coupling:** the inclusion indicator **destroys the ridge’s win** — `ssblasso` mean -0.312 , *worse* than the dense ridge -0.362 and $\approx$ the plain lasso. Selection discards the diffuse many-arm signal the dense ridge keeps. **Confirmed: you cannot have selection AND best coupling** (the spike-and-slab’s coupling loss, §2a, in the unified harness). - **Attribution/identifiability:** the inclusion **PIP is the BEST signal on the hard co-driver genes** — recovers ZEB1 miR-200/429 (2/2) and CDKN1A miR-15/16+miR-17~92 (2/2), which the dense ridge $|z|>2$ **misses entirely (0/2, 0/2)**: the evidence- π rescues evidence-supported drivers that pure data-driven identifiability can’t resolve under collinearity. But it *under*-recovers PTEN’s many drivers (2/4 vs 4/4) and the raw mean stays partly contaminated (un-informed edges leak on their included sweeps). - **The irreducible tension (now fully mapped):** coupling wants **dense** (no inclusion); identifiability on co-driver genes wants **inclusion** (evidence-informed). They pull opposite ways $\Rightarrow$ **no single fit is optimal for all three**. But it is the SAME Bayesian family — flip ONE knob: - $\pi \equiv 1$ (**dense**) $\rightarrow$ coupling (§2d winner) + attribution point estimate ($|z|>2$, §2e); - **evidence- π (inclusion)** $\rightarrow$ identifiability, esp. the hard co-driver genes (PIP >0.5). **One framework, two π -settings — not one fit.** That is the honest, defensible consolidation of the arc: a single learned- τ^2 non-negative Bayesian model, run dense for coupling/attribution and inclusion-on for co-driver identifiability. Full estimator grid in `output/learned/shrinkage_compare.tsv` + `attribution_eb.run()`.

The full 2 $\times$ 2 (slab shape $\times$ selection), coupling on the hub — collapses to ONE axis:

	ridge slab (Gaussian)	lasso slab (Laplace)
dense ($\pi \equiv 1$)	<code>ebridge_nn</code> -0.362	<code>blasso</code> -0.363
inclusion (spike)	<code>ssridge</code> -0.304	<code>ssblasso</code> -0.312

Slab shape (ridge vs lasso) is **irrelevant** — rows are flat ($-0.362 \approx -0.363$; $-0.304 \approx -0.312$). Only **dense vs inclusion** matters, and inclusion *hurts* coupling (both spike variants drop below even the plain lasso -0.320). So the entire estimator space reduces to a single knob: **dense $\rightarrow$ coupling/attribution, inclusion $\rightarrow$ co-driver identifiability.** Gaussian-vs-Laplace and CV-vs-EB are second-order; selection is the axis.

§3 The model — why non-negative, why adaptive, why α barely matters

Non-negativity is not a convenience; it encodes the biology “miRNAs repress.” Without the $M \geq 0$ box, OLS/lasso will happily assign a positive coefficient to an anti-correlated-by-confounding arm and call it activation — noise wearing a sign. The box turns “explain the target” into “explain the *repression* of the target,” which is the only claim we can defend.

Why adaptive (Zou reparametrization) and not plain lasso. Plain lasso shrinks all edges equally, so it drops a well-curated edge and a random one on the same footing. The adaptive

penalty lets curated edges resist shrinkage — the prior earns its keep exactly here. Implementation is the standard trick: scale column m by w_m , run a *uniform*- α non-negative lasso, unscale $M = b \cdot w$. Nothing exotic; the “adaptive” behavior is entirely in the column scaling.

Why α is a sparsity knob, not a coupling knob. Across $\alpha \in [0.001, 0.02]$ the mean gate ρ is flat (-0.340) and only the *count* of nonzero edges moves ($9 \rightarrow 7$); only $\alpha=0.05$ over-shrinks. So α is not a hyperparameter we agonize over — it tunes how many edges survive, not whether the aggregate couples. The canonical *attribution* (§5) sidesteps α entirely by being unpenalized bagged NNLS.

§4 Seed-family collapse — the estimand is family→gene

This is a statement about **what is identifiable**, not a modeling convenience. Same-seed arms share the targeting seed and are near-collinear in expression; no estimator can split “was it miR-200b or miR-200c” from bulk data — the design matrix simply lacks the variation. Rather than report a fabricated split, we **collapse the family into one predictor** and state the honest estimand: *this seed family → this gene*. The individual arm is then a **nomination**, not a claim. Pooling in true RPM space ($\log_2(1+\Sigma(2^x-1))$) and not in log space matters because abundances add in linear, not log, units. The family prior is the **max** member weight (the family is at least as evidenced as its best-studied arm).

§5 Canonical attribution — why bagged NNLS instead of the lasso’s coefficients

The single-lasso coefficient $M(m,g)$ is the **unstable functional** (bootstrap corr 0.03 under collinearity), so reading it directly to attribute a driver would report noise. Two fixes stack: **NNLS** (unpenalized, so no α -dependent shrinkage distorting the *relative* weights) and **bagging** (average over 40 bootstraps → variance down $\sim 1/B \rightarrow$ reproducible at corr ≈ 0.99). The **variance floor** ($sd < 0.1 \rightarrow$ zero weight) enforces a rule the biology demands: an arm undetectable-in-this-state cannot be a driver in this state, regardless of what the fit wants to do with its tiny residual variance. z-scoring is not cosmetic — level-scale NNLS is ill-conditioned and recovers high-variance noise arms; z-scoring puts every family on the same gauge so the weight reflects coupling, not dynamic range.

Why the same fixed support across states (§5’s ΔM). To compare a weight in tumour vs NAT vs GTEx, the *support* must be identical — otherwise “the weight changed” is confounded with “a different edge was selected.” Fixing the family support and re-fitting per state makes $dM = M_{state1} - M_{state2}$ a comparable estimand.

§5a Budget contribution — why family-size correction

A naive “ $M \cdot X$ ” would let a three-arm family count its abundance three times. Apportioning by each arm’s share of the family’s pooled abundance guarantees $\Sigma_{arm} contribution = M_{fam} \cdot X_{fam}$ — the family contributes once, split among its arms by realized abundance. A singleton is unchanged (its share is 1). This keeps “realized force” additive and honest at the family level.

§6 Validation gate — why out-of-fold, and what genome-wide FDR means

Why OOF and not in-sample fit. An in-sample $X \cdot M$ correlates with Y partly because it was *fit* to Y — that is circular and always “passes.” Holding out folds and scoring $X \cdot M_{train}$ on unseen patients makes the gate a genuine *predictive* test: the weights must generalize, not memorize.

Why three bars, not one. A significant coupling is not enough — it must beat the two baselines it could be a trivial restatement of. **Bar-1 (abundance):** if equal-weighting every arm couples just as well, the *learning* added nothing. **Bar-2 (curated fixed-M):** if freezing the prior weights couples just as well, the *fitting* added nothing (this is the heuristic the whole program nests and must improve on). Passing requires $\rho_{\text{model}} < \rho_{\text{abund}}$ **and** $\rho_{\text{model}} \leq \rho_{\text{curated}}$.

What “genome-wide FDR” actually means here. Each gene yields one pass/fail; testing $\sim 1,100$ – $1,500$ genes at $p < 0.05$ would produce dozens of chance “passes.” The genome-wide FDR (gate_fdr) computes, per gene, a one-sided partial-*t* p-value for $\rho_{\text{model}} < 0$ on $df \approx n - 8$ (the -8 = the confounder columns removed), then controls the **expected fraction of genes we call coupled that are actually null** across the whole scan. **Benjamini-Yekutieli, not just BH**, because the gene tests are **dependent through shared regulators** (one hub arm drives many genes); BY is valid under arbitrary dependence, BH would be anti-conservative. Caveat baked in: the test is **only meaningful for ≥ 2 -regulator genes** — for a singleton $X \cdot M \equiv \text{abundance up to scale}$, so there is nothing for “learning beats abundance” to detect.

§7 Per-edge coupling — what it adds over the gate, and why permute

The gate (§6) is an *aggregate* verdict on the whole regulator bundle. §7 zooms to **one (arm $\rightarrow$ gene) edge** and asks whether *that* arm’s abundance anti-correlates with the target after partialling out C and going to ranks (partial Spearman). This is the edge-level analog of the gene-level gate.

Why a permutation null instead of a textbook p-value — two independent reasons:

1. **Family dependence.** Same-seed arms are not independent tests. A parametric p-value assumes an independent-sample null and is miscalibrated. The permutation builds the null empirically and — crucially — applies the **same permutation π to every arm row**, so the null inherits the *same* cross-arm correlation structure as the observed statistic. Applies to apples. 2. **No clean analytic null for a partial-rank statistic.** After residualizing ranks on C, the sampling distribution of the partial Spearman is not a standard Spearman/t distribution. The permutation gives the exact finite-sample null for *this* statistic.

The two reported p-values do different jobs: p_{perm} is the raw permutation tail; $p_z = \Phi((\text{obs} - \mu)/\sigma)$ is a smooth Gaussian-tail approximation from the null moments, more stable in the extreme tail where few permutations land. Multiplicity (BH / BY / family-Simes) then runs across edges, with the **seed family as the hypothesis unit** (§14) so collinear arms are not counted as independent evidence.

§8 Causality (CN-locus instrument) — from correlation toward cause

§6–§7 establish *association*; they cannot rule out a third cause driving both arm and target. §8 reaches for causation with an **instrument**: the copy number of the miRNA’s own locus, $\text{CN}(m)$. The logic of IV — if a variable that moves the arm *only through the arm* (genetic dosage) also moves the target, that is much harder to explain by confounding than a plain correlation.

- **First stage / $F > 10$:** the instrument must actually move the arm; a weak instrument ($F \leq 10$) gives unreliable IV and is discarded.
- **Reduced form (ρ_{reduced}):** does residualized $\text{CN}(m)$ anti-correlate with residualized Y? This is the *genetic-dose* channel — expected < 0 (more locus copies $\rightarrow$ more arm $\rightarrow$ less target).
- **ρ_{obs} (the comparison baseline):** the plain observational partial-Spearman of the arm’s **measured abundance** vs the target, same C. It is the correlational channel. causal_concordant demands **both** $\rho_{\text{reduced}} < 0$ **and** $\rho_{\text{obs}} < 0$ (plus $F > 10$): the correlation you see is *corroborated* by the dosage channel. If $\rho_{\text{obs}} < 0$ but

$\rho_{\text{reduced}} \geq 0$, the observational anti-correlation is not backed by genetics $\rightarrow$ likely confounded, not causal.

Why cluster-exclusion is mandatory, not optional. CN moves the *whole genomic cluster*, so a co-located miRNA that also targets *g* violates the IV exclusion restriction — the instrument is no longer arm-specific. The fix conditions each candidate on its co-targeters: *arm_unique* requires the arm’s own conditioned partial to survive (< -0.1) while **no co-targeter** does. *strong_causal* = concordant AND (*cluster_clean* OR *arm_unique*) — a causal claim that survives the “was it really this arm or its neighbour” test.

§9 Driver resolution inside a family — honesty about identifiability

Given a family (§4) that the model selected, §9 asks *which member(s) carry the repression* by measuring each member’s anti-correlation **net of its family-mates** (+C). The design is built to **refuse to force a winner** when the data cannot support one — the three outcomes *are* the finding:

- **0 survivors $\rightarrow$ “family-only”**: the members are so collinear that conditioning removes each one’s residual variance (part $\rightarrow 0$); the split is genuinely unidentified (e.g. ESR1 miR-221/222). Reporting “miR-221 drives ESR1” here would be fabrication.
- **1 survivor $\rightarrow$ “single-driver”** (e.g. PTEN miR-181b).
- **≥ 2 survivors $\rightarrow$ “co-drivers”**: the repression is carried by several members, reported as a *set* (PTEN miR-17~92 = miR-106b + miR-20a; ZEB1 = miR-429 + miR-200c).

This is the guardrail that stops family collapse (§4) from silently becoming an over-claim about a single arm. The empirical ~13:7:11 single:co:family-only split says a large minority of “drivers” are honestly unresolvable — and the model says so.

§10 Uncertainty (hierarchical Bayes) — error bars, deliberately not point estimates

The lasso/NNLS give points, not posteriors, and fit each gene alone. §10 adds **calibrated uncertainty** by fitting a program-wise hierarchical Bayesian regression that pools each miRNA’s coefficient across all the genes it targets ($\beta(m, g) \sim N(\mu_m, \tau^2)$), yielding a posterior mean $\pm$ sd per edge and an “identified edge” rule $|\beta/\text{sd}| > 2$. Partial pooling shrinks noisy single-gene estimates toward the miRNA’s cross-gene behavior, so an edge reads as identified only if the data support it beyond its own uncertainty.

Two deliberate scope limits. (1) Cross-gene magnitude pooling does **not** improve OOF coupling here (the honest negative in VALIDATION §1) — so §10 is used purely as the **uncertainty object**, not as a better predictor; the coupling verdict stays per-gene (§6/§7). (2) It answers “how sure,” not “who” (§9) or “is it causal” (§8). Reading it as a driver-selector would overreach — its job is error bars.

§11 Subtype tests — three different questions, not one

“Does regulation differ by subtype” hides three separable questions, and conflating them is the classic error: **LEVEL** (does pressure magnitude $X \times M$ differ across PAM50 — Kruskal), **SLOPE** (does the *coupling* differ — a subtype $\times$ pressure interaction F-test), and **WIRING** (do the *weights* M themselves differ). A subtype can have higher pressure with identical wiring (level differs, wiring conserved) — so you must test the right one for the claim you want to make.

Why the pooled wiring estimator corrects the independent one. Fitting M *independently* per subtype differences two small-n noisy fits and **over-calls remodeling** — it flagged PTEN as rewired. The pooled estimator (Decision H) makes the whole-cohort M_{all} the **prior mean**

and estimates only a *regularized deviation* δ_s per subtype (ridge toward M_{all}), so small- n noise shrinks to zero instead of masquerading as rewiring. Result: PTEN is borderline ($\sim 1.35\times$ across λ , not robust), only RB1-Her2 modestly remodels $\Rightarrow$ **subtype wiring is mostly conserved**; the robust subtype signals are LEVEL and common-M SLOPE, not rewiring. This is the design's $M_T = M_H + \Delta$ nesting in its subtype flavour — the same shrink-the-delta logic that guards the state comparison.

§12 Discovery — why a new edge needs three gates, not one

An orphan (m,g) beyond the curated set must clear three independent hurdles, each blocking a different false positive: (1) **conditioned partial-Spearman** below threshold, conditioning on *C and the curated-HE aggregate* — so a “discovery” cannot be a shadow of the known regulators; (2) **scanMiR biochemical support** — a real duplex must be plausible, not just a correlation; (3) **composition retention** under deconv — the edge must survive stripping stromal/immune composition, i.e. be cell-intrinsic. Edge-level FDR comes from a permutation null-scan (real vs permuted hit counts), and hypotheses collapse to seed-family $\rightarrow$ gene (§14). The three gates are AND-ed because each alone has a characteristic failure mode (statistical shadow / no mechanism / composition artefact).

§13 Enrichment — a hypergeometric over the right universe

The only subtlety is the **universe**: enrichment of a target set in a Hallmark program is computed over the **HE universe** U , not all genes. Testing against the whole genome would inflate every fold-change (the HE set is already biased toward studied genes); conditioning on U asks the honest question — *given the edges we could have found, is this program over-represented*. BH across programs.

§14 Multiple-testing machinery — why the hypothesis unit is the family

The recurring principle: **the seed family is one hypothesis, not N arms**. Counting family-mate arms as independent tests would both inflate the test count (over-conservative multiplicity) and double-count correlated evidence (anti-conservative discovery). The toolkit matches the dependence structure: **BH** when tests are near-independent; **Benjamini-Yekutieli** (the $1/H_n$ factor) under arbitrary dependence, e.g. the gene-level gate where hubs induce dependence; **Simes-per-family** to roll family-mate p-values into one family q ; and a **family-wise min-statistic permutation null** for the permutation tests. Which one applies is dictated by *where the dependence is*, not by taste.

§15 Identity vs magnitude — two attribution objects that must never be conflated

Over a gene's active families there are two legitimate but *different* “who matters” answers, and reporting one as the other is a category error:

- **MAGNITUDE (budget)** = $M_{fam} \cdot X_{fam}$ normalized — *realized force*, abundance-included. Rewards an arm for being abundant.
- **IDENTITY** = **Shapley** credit for the fixed-weight aggregate's explained variance (LMG for the linear model) — *who explains the target*, abundance-removed. Rewards on-target coupling.

The gap = identity - magnitude is the interesting object: $+$ = a quiet on-target owner (explains more than its budget — ESR1 miR-18: identity 0.54 / budget 0.20), $-$ = a loud passenger (high budget, explains little — ESR1 miR-22: budget 0.26 / identity 0.01). The

gap is *primarily* abundance-vs-coupling and *secondarily* collinearity (co-varying families split shared R^2).

The discipline that keeps this from being abused: identity is $\perp$ **coupling** (§6/§7). The share *distributes the coupling that already exists*; it is **never** evidence that a driver exists. You establish existence with §6/§7/§8, then use identity only to apportion credit among established drivers. This retires the precursor softmax-share notion of identity, which blurred the two.

§16 Structural identity — the expression-free “who is designed to repress this”

Every object above needs the arm to be *expressed* to score. A specialist that is **silenced** in tumour has zero abundance, zero budget, zero Shapley — it vanishes from the realized picture even though it is the arm *designed* to own the gene. §16 recovers it with an **expression-free** score: biochemical potential (scanMiR/TargetScan strength $\times$ evidence) $\times$ specificity (the family’s pooled evidence-mass fraction aimed at g) $\times$ confidence (a depth multiplier — no evidence $\Rightarrow$ no ownership). Because there is **no biochemical specialist** (every arm hits ~ 700 targets roughly uniformly), specificity is a *pure evidence construct* — we are honest that “who specifically owns this gene” is a curation fact, not a biochemical one. Pooling the **whole seed family’s** evidence (not one representative arm) stops a thinly-studied member from faking concentration. `lost_specialist` overlays this with baseline-active-but-tumour-silenced to surface the arms a purely expression-based readout would miss.

§17 Loss/gain mechanism — why promoter methylation, and why bidirectional

If §16 says an arm is silenced, §17 asks *by what mechanism* — and promoter methylation is the testable one. The **no-cCRE-hop** detail matters: the gene-centric probe annotation misses miRNA loci (which is why the older pipeline fell back to cCREs), so §17 maps CpG probes **directly** over the pri-miRNA promoter window. **Bidirectional** because silencing is only half the story: hyper ($\Delta\beta$ up, arm silenced $\rightarrow$ repression *lost* $\rightarrow$ target up) and hypo ($\Delta\beta$ down, arm de-repressed $\rightarrow$ repression *gained* $\rightarrow$ target down) are symmetric mechanistic calls. The TSG-vs-oncomiR validation ($\Delta\beta$ +0.061 vs -0.070 ; miR-124 +0.193, miR-21 -0.176) confirms the direction convention points the right way.

Reading guide — which doc answers your question

You are asking...	Go to
What is the exact objective/formula?	LEARNED_MODEL_METHODS.md (§ same number)
Why is it built this way / what breaks otherwise?	this doc (§ same number)
Which estimator do I use for job X, and vs what?	LEARNED_MODEL_ESTIMATOR_MAP.md
Does it work — numbers, controls, honest negatives?	LEARNED_MODEL_VALIDATION.md
Why was the whole thing designed this way?	LEARNED_MODEL_DESIGN_RESPONSE.md

Every §here mirrors the same § in METHODS. If they ever drift, METHODS (the spec) is authoritative for formulas and this doc is authoritative for reasoning — reconcile, don’t let them diverge.